# V SADANANDAM M.Tech(Aerospace Engineering)IIT-Chennai
B.Tech(ECE)JNTU-HYD, Diploma(Chemical)

Chief Manager (Design)–Avionics System Design-SW-HW&IVVQ

Strategic Electronics Research & Design Centre (SLRDC)

Hindustan Aeronautics Ltd (HAL), DPSU under Ministry of Defence (GOI), Hyderabad-India

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Raksha Mantri Award holder from MOD(GOI)-2022
Leading R&D/System Engineering Projects 20years+
Leading Primary, Secondary and Doppler RADARs
DSP algo-FIR,1D/2D/3D-FFT, CA-CFAR, Kalman_filter
Lead & develop embedded, RF & DSP based systems
Expertise in A&D standards and regulations
Leading global teams UK, RUSSIA and FRANCE
PMP certification from GOOGLE
Platform integration experience (Fixed/Rotary wing)
Patent holder -GOI &Patent applications progress
SDLC with IVVQ expertise 15years+ DO178B, DO254
DIGITAL TECHNOLOGIES 8years+, AI tools 2years+
Process Automation, Digital twin,Dig-thread, IoT
Expertise in Desing thinking, LEAN, Six sigma+CI
Managing cross functional teams 15years+
Lead member PLM-Team center, ERP tools
PMP and Coach/Tutor on aircraft/avionics systems
Multi-disciplinary core expertise(HW,SW,RF,System Design)
SE concepts SWAP-C with AI tools (rapid proto, automation)
SE Program skills C,VHDL,PYTHON, MATLAB, MBSE
Excellent Interpersonal and Communication skills
Safety Assessment process ARP4754A, ARP4761
SIOP (Sales, Inventory, Operations, Production)
Highly Energetic- Agile-Focused, Dedicated and Dynamic

Web: https://www.linkedin.com/in/sadanandam-aersopace-defence

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EXECUTIVE PROFILE

Strategic Engineering Executive with **20years+ of leadership** in the aerospace and defense R&D industry. Expert in the design, development and airworthiness certification of safety-critical avionics including **RADARS (Primary/Secondary/Doppler RADARs), NAVIGATION (DVS/INS/GNSS systems), CDU (Controller/Display Units), and Knowledge of EW SUITE, DATA LINK and iCNS systems** with proven track record of **RAKSHA-MANTRI** award (innovation category-2022) and **Patent** on advanced DSP algo and steering large-scale engineering departments, managing multi-million dollar R&D budgets, and securing **CEMILAC-Type Approvals, DGCA-ITSOA certifications** for fitment on wide variety of aircrafts (Fixed wing and Rotary wing) and Missile platforms.

CORE COMPETENCIES

Strategic Design Expertise with Technology Reviews: Design & Development of various RADARs (Primary/Secondary/Doppler) / NAVIGATION (DVS/INS/GNSS systems) / CDU (Controller/Display units)with overall system architecture finalization reviews - SRR, PDR, CDR MRR and conducting Software reviews (SOI#1 to SOI#4), Multi-disciplinary Team Management, R&D Road mapping, Various Project proposals with Corporate office/MoD/Government Liaison and complete life cycle management of avionics systems from design to deployment with knowledge of EW SUITE and DATA LINK Systems.

RADAR Systems: Radar Altimeters, RPF and ARPF (Altimeter cum RPF) systems, IFF Mark XII, Combined Interrogator Transponders (CIT).

Navigation Suite: Doppler Velocity Sensors (DVS), GNSS, INS, Integrated GNSS/INS (EGI) and TACAN systems.

CDU (Controller Display Units): LCD/AMLCD displays Controllers/Simulators for various avionics interfaces ARINC429 and MIL-STD-1553B etc

System Software development: Application software development using DSP algorithms, embedded software development for avionics digital interfaces/controllers

System Hardware development: Hardware design of various modules for RADAR and NAVIGATION systems, AVIONICS interfaces, controllers/simulators etc

Avionics Digital/Analog interfaces: ARINC429/MIL-STD 1553B/ARINC 664(AFDX) digital interfaces, ARINC-554 for analog interface etc

Defence Standards: MIL-STD-810, MIL-STD-461, MIL-STD-1553B, ARINC 664 (AFDX) and ARINC429, DO-178C (DAL A/B), DO-254, ARP4754A, ARP4761

Civil Standards: DO-160, ARP4754A, ARP4761, DO-178C (DAL A/B), DO-254, MIL-STD-1553B, ARINC 664 (AFDX) and ARINC429

CEMILAC/DGCA-GOI Certifications: RCMA/CEMILAC–Type Approvals of LRUs, SOF/ESS clearances for LRUs, PCs and DGCA/ITSO/STC certification process.

DSP algorithms: Lead & develop advanced DSP algorithms to analyze, processing down-converted RF/IF signals in Rx chain: ADC sampling, FIFO, FIR filtering, 1D/2D/3D-FFT Spectrum analysis, CA-CFAR algorithms for Fixed Pd and PEAK detection algorithms etc.

Aircraft integration: Successful Integration and satisfactory performance proven track record of variety of avionics viz., RADARs / NAV SUITE / CDUs on various aircrafts.

PLM and SDLC with IVVQ expertise 15years+

SIOP: Sales, inventory and Operations planning expertise 10years+ with excellent resource planning and allocation.

ENGINEERING R&D EXPERIENCE

Chief Manager (Design) – Avionics Systems Design & Integration

[SLRDC, Hindustan Aeronautics Ltd (HAL), DPSU under MoD(GOI)- Hyderabad

Career profile				
Grade	**Designation**	**Division**	**From**	**To**
VI	**Chief Mgr(Design)**	SLRDC, HAL-HYD*	01/07/2022	present
V	**Sr Mgr(Design)**	SLRDC, HAL-HYD	01/07/2018	30/06/2022
IV	**Manager(Design)**	SLRDC, HAL-HYD	01/01/2014	30/06/2018
III	**Dy Mgr(Design)**	SLRDC, HAL-HYD	01/01/2010	31/12/2013
II	**Engineer(Aero)**	SLRDC, HAL-HYD	17/10/2006	31/12/2009
DT	**Engineer (Design Trainee)**	SLRDC, HAL-HYD HMA-Bangalore	15/10/2005	16/10/2006
* SLRDC - **S**trategic E**l**ectronics **R**esearch & **D**esign **C**entre HAL - **H**industan **A**eronautics **L**imited HYD - **Hy**derabad				

Role: Execution of various leadership roles in Design, development and certification of various Avionics systems (Hardware, Software, System architectures) for a leading aerospace entity. Responsible for the technical road-map, system design readiness, progress-review meetings with various teams and coordination, conducting SRR, PDR, CDR, MRR, FTR reviews, integration of systems and coordination of flight trials with flight trial data analysis, conducting Software reviews, coordination of various stake holder viz., designers/engineers team, quality and safety management teams, airworthiness certification teams (RCMA/CEMILAC reps) and CRI teams (ORDAQA customer reps) and certification of mission-critical RADAR Systems, NAVIGATION SUITE (DVS+INS/GPS sensors) and CDU systems.

AWARDS, PATENTS and RECOGNITIONS

Received **RAKSHA MANTRIS award** for excellence under the category **RAKSHA ANVESHAN RATN (2021-2022)** for proving the **Radar Altimeter system design** on **SU30, Jaguar** and **LCA** fighter aircrafts with 1% accuracy up to an altitude of 20,000ft (6000m).

Received **patent from Patent office-GOI (Patent no. 414710, application no. 4082/CHE/2014, Date of filing 21/08/2014** for design and development of **Digital Radar Altimeter system** with **Advanced Digital Signal Processing algorithms.**

Patent filed in year 2024 on **ATE(Automatic Test Equipment) system development** for Radar Altimeter sensor.

Patent filed in year 2024 on **Nano-Radar Altimeter** sensor development for Missile applications -**DDC based PLL circuitry for generating FMCW signal in C-band and DSP algorithms for altitude computation of aircraft with range resolution of 0.5m.**

Received 1st rank in B.Tech (Electronics & Communication Engineering) from JNTU-Hyderabad university with 75%

Received 1st rank in Diploma in Chemical engineering (Petrochemicals specialization) from GIPDCE&T VIZAG with 82%

Received 1st rank in 10th class -Secondary School certificate with 83.3%

Received appreciations from various forums and FltOps/Chief test Pilots (Flt Operations Groups- HAL, NFTC) for proving various systems on SU30, LCA MK1, JAGUAR DIII fighter aircrafts (Fixed wing fighter platforms)

Received appreciations from various forums and FltOps/Chief test Pilots (Flt Operations Groups- Rotary wing) for proving various systems on ALH, LCH, LUH aircrafts (Rotary wing platforms)

Received appreciations from various forums and FltOps/Chief test Pilots (Flt Operations Groups- Rotary wing) for proving various systems on HAWKi, HJT36 aircrafts (Fixed wing training platforms)

EDUCATION- CREDENTIALS

M.Tech (Aerospace Engineering) –with expected **CGPA 8** [IIT Chennai] [2023 to 2026]

B.Tech (Electronics and Communication Engg) with **75%**[JNTU-Hyd] [2001 to 2005]

Diploma Chemical engg (Petrochemicals) with **82%** [GIPDCE&T–Vizag] [1997 to 2000]

SSC (**Siddhartha Residential School)** with **83.33%** [Karimnagar] [1995 to 1996]

ENGINEERING PROFICIENCY

Design/Simulation: VDSP++ IDDE, KEIL IDE, XILINX ISE/VIVADO, MATLAB, Simu-link, Keysight ADS (RF), HFSS, ORCAD, PYTHON AI+ML and System-vue tools, Excel with AI, Prompt Engineering with AI, Power BI tools, RTOS, MBSD (Model Based System Design) with MATLAB and VIVADO tools

Languages: C, C++, JAVA, C#, PYTHON and VHDL

Life-cycle: SIEMENS TEAM CENTRE [PLM with integrated POLARION (ALM tool)] LDRA, ALDEC, IBM RATIONALE DOORS

Reliability: RELEX (MIL-HDBK-217)

Business/AI tools: MS-Office, Power-BI, Agen-AI (n8n) and Claude/LMNotebook

ERP: SDLC Team Center(PLM & ALM), IFS/ERP

VALUES AND SKILLS

INNOVATIVE THINKING

LEADING BY EXAMPLE

AGILE AND ADAPTABILITY

CREATIVITY WITH CRITICAL THINKING

TEAM WORK

DEVELOPING PROCESSES WITH CONTINUOUS IMPROVEMENT

PROACTIVE AND RESULT ORIENTED

ENGINEERING EXPERTISE-SYSTEM DESIGN-DSP ALGORITHMs-EMBEDDED CODE

Expertise in complete system design - FMCW Radars, Doppler Radars, Primary and Secondary Radars, Navigation Systems (Nav Suite), Radio Proximity Fuses, Collission Avoidance systems, Simulators, Controllers with various DSP algorithms, Embedded code developments and GUI development etc

Expertise in system design architecture viz., Integrated Modular Architecture System design (IMA) concept and Stack-based architecture to achieve low-SWaP-C (size, weight and power-cost).

Expertise in complete system software package development viz., application software code development for various avionics systems, embedded code development for various avionics interfaces,

Expertise in model based system design development and Complex electronic hardware development for various RADAR/NAV systems etc using various IDDEs (refer Technical Proficiency cited above)

Expertise in system hardware development - RF-analog-DIGITAL-power supply - Avionics interfaces(digital and analog) and various discrete inputs/outputs etc

Expertise in RF-power budget analysis for RADAR/NAV systems

Expertise in Schematic design of various analog/digital/interface/PS modules and knowledge of PCB design with SI analysis, thermal analysis and PI analysis etc

Expertise in developing AVIONICS digital interfaces viz, MIL-STD-1553B, ARINC429 and AFDX protocols etc.

RADAR receiver signal processing algorithms for detection of Fb(beat frequency)- analog down conversion (RF-base band), ADC sampling, windowing, Digital-FIR filtering, FFT processor for estimating Fb and altitude of the aircraft.

RADAR receiver signal processing algorithms for detection of Fd (doppler signals-4 channel DVS system time-shared)- 1st stage analog down conversion (RF-IF band), ADC sampling, windowing, 2nd stage DDC [Digital down conversion (IF to base band)], Digital-FIR filtering, FFT processor for 1Hz resolution of Fd (doppler signals).

RADAR receiver signal processing using Janus centroid algorithms using ZOOM-FFT processor and centroid/parabola interpolation techniques for detection of doppler singals with 0.125Hz resolution and 3D velocity tracking of aircrafts

RADAR receiver signal processing algorithms using model-based-development-approach using MATLAB and VIVADO tools

RADAR receiver tracking algorithm with KALMAN_FILTER for fine tuning of the various outputs

RADAR receiver tracking algorithms for detection of multiple targets

RADAR receiver tracking algorithms with advanced DSP techniques WELCH algorithm consisting of FFT processors and integration of several frames

RADAR receiver tracking algorithms for range, velocity and angle measurements

RADAR receiver tracking algorithm-adaptive threshold methodology using CA-CFAR techniques for detection of weak signals (terrain reflected) with varying SNRs

Embedded software code development for ARINC429 protocol

Embedded software code development for MIL-STD 1553B protocol

Embedded software code development for LCD/AMLCD controller/simulators

Embedded software code development for RS422/RS232/USB interface

GUI development using C# for ARINC429/MIL-STD-1553B/RS232Bprotocols

GUI development using C# for various ATE systems (RADAR ALTIMETER, TACAN and RPF systems)

Knowledge of ALT-8000 Simulator (VIAVI systems, singapore)

Knowledge of RADAR TARGET SIMULATOR using ODL (Optical Delay Line) technique

Knowledge of RADAR TARGET SIMULATOR using DRFM (Digital Radio Frequency memory) technique

Knowledge of python tool - AI/ML programs

Knowledge of KALMAN FILTER and Fusion algorithms

Knowledge of Power supply design wrt MIL-STD-704D compliance

Knowledge of RF front end design for RADAR/NAVIGATION systems wrt MIL-STD-461E compliance

Knowledge of various ARINC standards ARINC-552A, ARINC707-7, ARINC429

Knowledge of Schematic design, PCB design with SI analysis, thermal analysis and PI analysis etc

Knowledge of various DEFENCE standards MIL-STD-810, MIL-STD 461, MIL-STD 704D and MIL-STD 1553B

Knowledge of various CIVIL standards ARP-4754A, ARP-4761, RTCA-DO-160, TSO, RTCO-DO178B/C and RTCA-DO-254

Knowledge of mechanical CAD tools for avionics system design, thermal management via conduction/convection/radiation methods

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AVIONICS PROJECTS (RADAR, NAVIGTATION AND DISPLAY SYSTEMS)

Slno	Name of the Project	Model	Platform	Year
1	DIGITAL RADIO ALTIMETER RANGE UP TO 5000ft with 2% accuracy	RADALT_5000	ALH-MK1 LCH CHETAK	2010-2020
2	DIGITAL RADIO ALTIMETER RANGE UP TO 20,000ft with 1% accuracy	RADALT_20000	JAGUAR D-III SU30 UPG LCA DORNIER	2013-2025
3	RADALT DISPLAY UNIT-ARINC429	RAMDI-ARINC	GTE	2010-2012
4	RADALT DISPLAY UNIT-1553B	RAMDI-1553B	GTE	2012-2013

5	RADIO ALTIMETER DISPLAY UNIT (AMLCD)	RAMDU	CHETAK	2019-2022
6	TACAN AIR NAVIGATION	TACAN	SU30UPG DORNIER	2014-2020
7	IDENITIFICATION OF FRIEND OR FOE	IFF 2410	ALH LCH LUH	2010-2025
8	RADIO PROXIMITY FUSE	RPF_MISSILE	ASTRA MISSILE	2015 – 2018
9	ALTIMETER+RPF	ARPF_MISSILE	VLRSAM MISSILE	2018 - 2022
10	COMBINED INTERROGATOR TRNASPONDER	CIT_AIRBORN	LCA SU30 DORNIER	2012 - 2020
11	ALTIMETER+DISPLAY UNIT	RADALT+RAMDU	AN32 ALH MK1	2023- 2025
12	INSITU PROGRAMMING FEATURE FOR VARIOUS ALTIMETER MODELS	INSITU	LCA/DORNIER KUAV, UHM	2015 2023
13	LIGHT WEIGHT RADIO ALTIMETER (ITSOA Certification)	RADALT_CVL1	ALH CIVIL DORNIER CIVIL	**UNDER DEVELOPMENT (2020 onwards)**
14	LIGHT WEIGHT RADIO ALTIMETER (CEMILAC Certification)	RADALT _LIGHT	KIRAN-UAV UHM	**UNDER DEVELOPMENT (2021 onwards)**
15	COMPACT RADIO ALITMETER (MISSILES) DEVELOPMENT	COMP_RADALT	MISSILEs/UAVs	**UNDER DEVELOPMENT (2023 onwards)**
16	SINGLE ANTENNA RADAR ALTIEMTER	SARA_ALT	MISSILEs/UAVs	**UNDER DEVELOPMENT (2024 onwards)**
17	COMPACT mmWave RADAR ALTIMETER (24GHz and 77GHz)	COMPACT-RALT	DRONES	**UNDER DEVELOPMENT (2025 onwards)**
18	TRAFFIC ALERT AND COLLISION AVOIDANCE SYSTEM	HTACAN	MIL/CIVIL	**UNDER DEVELOPMENT (2025 onwards)**
19	DOPPLER VELOCITY SENSOR (DVS) DEVELOPMENT	NAV SUITE	ALH LCH UHM MI17	**UNDER DEVELOPMENT (2025 onwards)**
20	INS SENSOR DEVELOPMENT			
21	GNSS SYSTEM DEVELOPMENT			

ENGINEERING CHALLENGES/SNAGS ADDRESSED

Slno	Name of the Project	Model	Platform	Methodology
1	Addressed the issue of intermittent search problem on RADAR ALTIMETER units	RAM 1750A	ALH	SEARCH-TRACK ALGORITHM IMPROVEEMNT
2	Addressed the RADAR ALTIMETER leakage issue between Tx and Rx antennae at higher bank angles	RAM 1760A	LCH	TRACKING ALGORITHM IMPROVEMENT – 2ND THRESHOLD METHODLOGY INTRODUCED TO ISOLATE THE LEAKAGE
3	Improved the range resolution of RADAR ALTIMETER units in steps of 1ft up to 20ft o	RAM 1750A ALN1	ALH-MKIII NAVY	TRACKING ALGORITHM IMPROVEMENT – PARABOLIC INTERPOLATION TECHNIQUES ADAPTED TO ESTIMATE THE ACTUAL PEAK BIN IN THE DIGITAL SPECTRUM OUTPUT
4	Addressed the issue of latency in RADALT height output from 1sec to 300msec	RAM1780A	CHETAK	TRACKING ALGORITHM IMPROVEMENT – MOVING AVERAGE ALGORITHM IS FINE TUNED FOR FASTER UDPATE RATE
5	Addressed the issue of intermittent search of RADALT units at lower heights	RAM 1770A	LUH	TRACKING ALGORITHM IMPROVEMENT – USING CA-CFAR ADAPTIVE THRESHOLDING ALGORITHM
6	Addressed the issue of latency in ADVANCED RADAR ALTIEMTER units –height output from 1.2 sec to 300msec	RAM2700A	SU30	TRACKING ALGORITHM IMPROVEMENT –KALMAN FILTER ALGORITHM GAIN FACTOR IS FINE TUNED FOR FASTER UPDATE RATE AND RECEIVED **RASKSHA MANTRIS AWARD (2022)** ON SATISFACTORY FLIGHT TRIALS
7	Aircraft ground Integration Issues	RAM2700A	LCA DORNIER	ISOLATED THE CABLE LOOM ISSUES
8	Aircraft ground Integration Issues	RAM1750A ALN1	SEAKING	ISOLATED THE CABLE LOOM ISSUES wrt GROUN SIGNALS
9	Aircraft ground Integration Issues	TACAN	SU30	ISOLATED THE CABLE LOOM ISSUES wrt INTERFACE
10	Aircraft ground Integration Issues	IFF	ALH	ISOLATED THE CABLE LOOM ISSUES
11	Aircraft ground Integration Issues	CIT	DORNIER	ISOLATED THE CABLE LOOM ISSUES
12	Successfully completed the ground integration checks of Light weight RADAR ALTIMETER units	RAM1770A	KIRAN-UAV and UHM platform	Flight trials in progress
13	Successfully completed the ground integration checks of RADAR ALTIMETER units	RAM1750A	AN-32 and ALH MK-1 platform	Flight trials in progress

PROFESSIONAL CREDENTIALS

Various training and certifications programs on professional competency, leadership skills, DGCA/CEMILAC certifications and life cycle management of avionics systems from various organizations HMA (Hindustan Management Academy) Bangalore, TTC (Technical training center)-HAL-HYD, JNTU-HYD and IIT-Chennai etc

Training profile			
Training program	Institution	From	To
Certificate on AI Tools workshop	Be10X	25-Jan-2026	25-Jan-2026
SAARATHI: Leading the future for Design Complex	HMA-Bangalore	05-May-2025	08-May-2025
Work-Life balance & Mental health being	HAL-HYD	09-Apr-2025	09-Apr-2025
CAR-21Design Organization Approval (DOA)	HAL-HYD	11-Nov-2024	11-Nov-2024
Overview of Model CDA Rules	HAL-HYD	13-July-2024	13-Jul-2024
Mentoring Skills: PARAMARSH	HMA-Bangalore	20-May-2024	22-May-2024
Thinking Fast Thinking Slow	HMA-Bangalore	01-Feb-2024	03-Feb-2024
HAR GHAR DHYAN (OHS)	HAL-HYD	23-Sep-2023	23-Sep-2023
TRAINING ON CAR-21 & DOM	SLRDC-HYD	29-Aug-2023	29-Aug-2023
EMOTIONAL INTELLIGENCE & HOLISTIC WELL BEING	TTC-HYD	13-Dec-2022	15-Dec-2022
SYSTEM SAFETY ASSESSMENT	HAL-HYD	21-Nov-2022	21-Nov-2022
CAR-21 AND DOM TRAINING	HAL-HYD	19-Oct-2022	19-Oct-2022
AWARENESS SESSION ON ISO27001:2013	HAL-HYD	06-Sep-2022	06-Sep-2022
INDUSTRY 4.0 (ONLINE)	HMA-Bangalore	02-Feb-2022	25-Feb-2022
ML & AI : FOUNDATION WORKSHOP (ONLINE)	HMA-Bangalore	20-Dec-2021	21-Dec-2021
AI FOR EVERYONE: MASTER THE BASIC	HMA-Bangalore	29-Nov-2021	27-Dec-2021
AIRCRAFT SYSTEMS (ONLINE)	HMA-Bangalore	04-Oct-2021	28-Oct-2021
CAR-21	HAL-HYD	18-Nov-2019	19-Nov-2019
SCIENCE OF PRACTICE OF HAPPINESS	HMA-Bangalore	10-Apr-2019	12-Apr-2019
AWARENESS SESSION ON AS9100D	HAL-HYD	27-Oct-2017	27-Oct-2017
INTERNAL AUDITORS-ISO 50001	HAL-HYD	18-Dec-2013	19-Dec-2013
CON TITLED ON ICRAMAV -2013	JNTU-HYD	21-Nov-2013	23-Nov-2013
DT REFRESHER PROGRAMME	TTC-HYD	19-Sep-2011	24-Sep-2011
JOB DESCRIPTION WORKSHOP	TTC-HYD	27-May-2008	27-May-2008

PROFESSIONAL TRAITS:

Professional Competence:
Functional managerial ability/ and attitude in application of job knowledge to set perspective plan & fulfillment of the same with optimum use of resources.

Cost & Time Consciousness:
Utilization of resources effectively, ensuring required tolerance / quality within optimum time and minimum cost.

Developing Subordinates:
Ability to interact, guide, council & nurture subordinates with a view not only to help them to perform their present job effectively but also to groom them for meeting higher responsibility / challenge.

Team Building:
Ability to interact, command & influence people with a view to have cohesiveness amongst them to ensure thrust in the direction of achievement of targets.

Potential to shoulder Higher responsibilities:
Ability to take -up higher level responsibility and set pace for the team members towards goal achievement.

AREAS OF INTEREST:

Leading the R&D/System Engineering projects/manufacturing with strategic vision

Adapting Emerging Technologies in System Engineering-Avionics/Aerospace domain

RADAR technologies (SWaP-C) + AI based algorithms

Navigation suite sensors (AI Quantum Nav + INS/GPS + video based Nav)

Flight Management systems (Sensors+Autopilot+Displays) + AI based algorithms

Embedded systems- ARM+FPGA based

Aircrafts/Helicopter technologies R&D/manufacturing with foreign OEMs

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REFERENCES:

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Government of India
Ministry of Defence

Department of
Defence Production

Azadi Ka Amrit Mahotsav

Raksha Mantri's Award For Excellence

Raksha Anveshan Ratn
(2021-2022)

Awarded to

Shri Thiyagarajan Viswanathan & Team
Strategic Electronics Research and Design Centre ,
Hindustan Aeronautics Limited

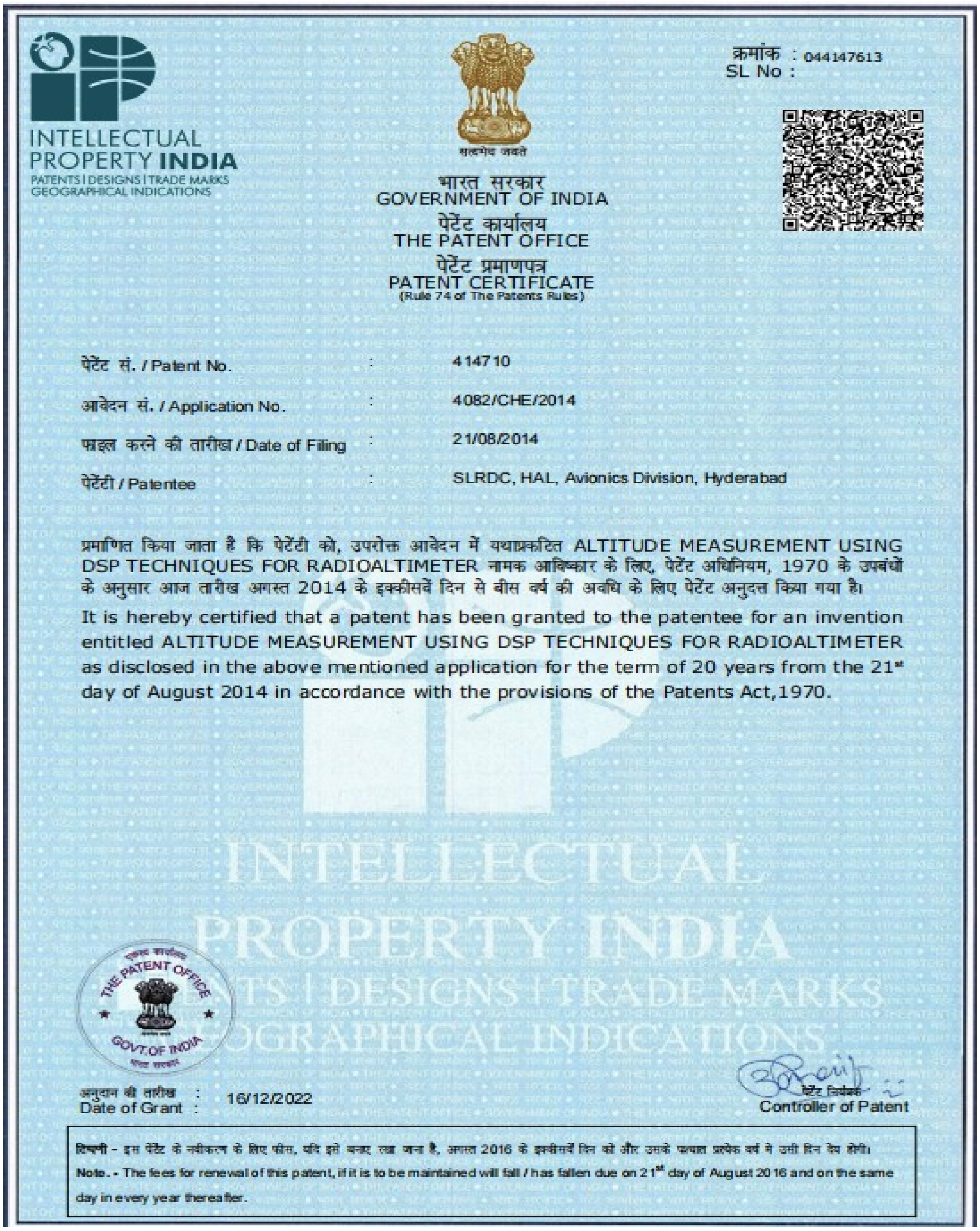
Innovation / Technological Breakthrough
Under Category – Individual / Team

(Dr. Ajay Kumar)
Defence Secretary

(Shri. Rajnath Singh)
Raksha Mantri

20th October, 2022. Gandhinagar, Gujarat

RAKSHA MANTRI AWARD FOR EXCELLENCE



क्रमांक : 044147613
SL No :

INTELLECTUAL PROPERTY INDIA
PATENTS | DESIGNS | TRADE MARKS
GEOGRAPHICAL INDICATIONS

भारत सरकार
GOVERNMENT OF INDIA
पेटेंट कार्यालय
THE PATENT OFFICE
पेटेंट प्रमाणपत्र
PATENT CERTIFICATE
(Rule 74 of The Patents Rules)

पेटेंट सं. / Patent No.	:	414710
आवेदन सं. / Application No.	:	4082/CHE/2014
फाइल करने की तारीख / Date of Filing	:	21/08/2014
पेटेंटी / Patentee	:	SLRDC, HAL, Avionics Division, Hyderabad

प्रमाणित किया जाता है कि पेटेंटी को, उपरोक्त आवेदन में यथाप्रकटित ALTITUDE MEASUREMENT USING DSP TECHNIQUES FOR RADIOALTIMETER नामक आविष्कार के लिए, पेटेंट अधिनियम, 1970 के उपबंधों के अनुसार आज तारीख अगस्त 2014 के इक्कीसवें दिन से बीस वर्ष की अवधि के लिए पेटेंट अनुदत्त किया गया है।

It is hereby certified that a patent has been granted to the patentee for an invention entitled ALTITUDE MEASUREMENT USING DSP TECHNIQUES FOR RADIOALTIMETER as disclosed in the above mentioned application for the term of 20 years from the 21st day of August 2014 in accordance with the provisions of the Patents Act,1970.

अनुदान की तारीख :
Date of Grant : 16/12/2022

पेटेंट नियंत्रक
Controller of Patent

टिप्पणी - इस पेटेंट के नवीकरण के लिए फीस, यदि इसे बनाए रखा जाना है, अगस्त 2016 के इक्कीसवें दिन को और उसके पश्चात प्रत्येक वर्ष मे उसी दिन देय होगी।
Note. - The fees for renewal of this patent, if it is to be maintained will fall / has fallen due on 21st day of August 2016 and on the same day in every year thereafter.

PATENT ON ADV DSP ALGORITHMS FOR RADIO ALTIMETER